

Clinical Findings

Actinomycosis should always be suspected when one of the following three features is present: a mass-like inflammatory infiltrate of the skin and subcutaneous tissue, sinus tract formation with drainage, or a relapsing or refractory clinical course after short-term antibiotic therapy. Actinomycotic granules may be visible macroscopically.

Cervicofacial Actinomycosis

Cervicofacial actinomycosis is the most common form, accounting for approximately 55% of cases. Predisposing factors include poor dental hygiene, dental or periodontal disease, recent dental procedures, oral surgery, or penetrating trauma to the oral mucosa. Infections often originate as a periapical abscess.

The most frequent location is the angle of the jaw and high cervical region (60%), followed by the cheek (16%), chin (13%), and less commonly, the temporo-mandibular joint and retromandibular area. The lesion typically begins as a firm, indurated mass that may mimic a neoplasm. Over time, it can progress to recurrent abscesses and spread across tissue planes without regard for anatomical boundaries.

Lymph node involvement is rare, but extension to adjacent structures may occur, including the orbit, cranium, spine, and vascular structures. The lesion is often painless, though pain, fever, and leukocytosis may be present. As the infection reaches the skin surface, sinus tracts develop, and the overlying skin may appear purplish-red. Trismus can occur due to muscular involvement.

Bone involvement occurs in about 10% of cases, most commonly affecting the mandible. Limited disease may present as an oral or nasopharyngeal mass or ulcer. Extension to the ear may manifest as chronic otitis media or mastoiditis. Ocular complications include orbital cellulitis, lacrimal canaliculitis, and endophthalmitis.

Thoracic Actinomycosis

Thoracic actinomycosis accounts for approximately 15% of cases. The primary route of infection is aspiration of oropharyngeal organisms, although spread from cervicofacial disease to the mediastinum can also occur. The infection may involve the lung parenchyma, pleura, mediastinum, and chest wall.

The clinical course is indolent, with symptoms including chest pain, fever, weight loss, cough, and occasionally hemoptysis—often mimicking tuberculosis. Radiologically, it may appear as a pulmonary mass or pneumonia, frequently with pleural extension.

Cutaneous involvement of the chest wall occurs in up to 26% of thoracic cases, presenting as abscesses and sinus tracts. Concurrent involvement of lung parenchyma, pleura, and chest wall strongly suggests actinomycosis. Mediastinal disease may present as anterior chest wall abscesses (rarely involving the sternum) or paraspinal abscesses. Breast tissue and breast implants can also be affected.

Abdominal Actinomycosis

Abdominal actinomycosis represents about 20% of all cases. It typically arises from direct extension of gastrointestinal or genitourinary pathology, often following conditions such as appendicitis, diverticulitis, surgery, or trauma. The infection spreads contiguously through the abdominal wall.

Perineal and Genital Involvement

Perineal disease results from pelvic organ involvement, frequently associated with the use of intravaginal devices or intrauterine devices (IUDs). Infection spreads from the genital tract to surrounding soft tissues.

Hand Involvement

The only exception to endogenous infection is hand actinomycosis, which follows penetrating trauma such as fist bites or other inoculation events.

Histopathology

In tissue, the bacteria form filamentous aggregates known as sulfur granules (see Table 185-1). These are typically surrounded by neutrophils, chronic inflammatory cells, granulation tissue, and fibrosis. Granuloma formation is uncommon. Fistulas may allow drainage of sulfur granules, but their presence alone is not pathognomonic for actinomycosis.